CatGuard

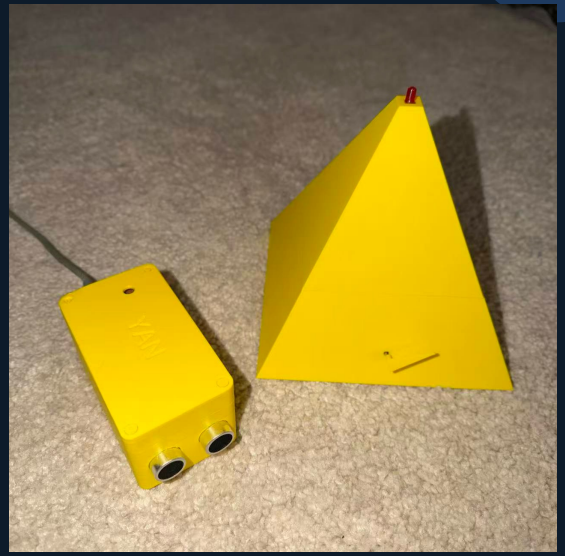
Distributed Ultrasonic Cat Detection & Deterrence System

TECH 514 • Final Project • Winter 2026

Yan Zhang

GitHub: github.com/yanzhang1103/TECH514finalproject

"Cat enters — system responds. Cat leaves — silence resumes."



The Problem

🐱 Curious Cats

Cats instinctively explore and scratch valuable objects — keyboards, houseplants, fragile items, or restricted zones — with no regard for boundaries.

⚠️ No Real-Time Detection

Existing solutions (physical barriers, sprays) require constant human presence or manual reset. There is no smart, automated proximity monitoring.

📡 Single-Point Limitation

Most commercial deterrents are single-unit devices. Multi-room or complex layouts require independent, coordinated sensor nodes.

🔔 Delayed Response

By the time a human notices the intrusion, damage may already be done. Millisecond-scale automated deterrence is needed.

Proposed Solution

CatGuard is a two-device, distributed IoT deterrence system. A Sender device continuously monitors proximity using ultrasonic sensing; when a cat intrudes, it wirelessly triggers a Receiver node to activate mechanical, auditory, and visual deterrents — all within 80ms.

SAFE	WARN	ALERT
> 150 cm	100–150 cm	< 100 cm
Motor reset All alerts OFF	Motor → 90° Passive alert	Motor → 180° Buzzer + LED ON

System Architecture

📡 SENDER device

XIAO ESP32-C3

- HC-SR04 Ultrasonic Sensor
- Auto-calibration (5s boot)
- Sampling interval: 50ms
- USB-C powered (24/7)

ESP-NOW

< 80ms latency
No router needed

🔔 RECEIVER device

XIAO ESP32-C3

- X27 Stepper Motor (torque-compensated)
- Passive Buzzer (2.7kHz resonance)
- High-brightness LED
- 3.7V 820mAh LiPo battery

Critical Components & PCB

XIAO ESP32-C3

Main MCU (×2)

HC-SR04

Ultrasonic Sensor

X27 Stepper Motor

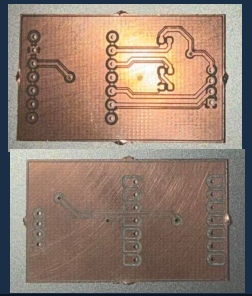
Passive Buzzer

Auditory Alert

High-brightness LED

Visual Alert

Custom PCB



Signal Processing & Adaptive Calibration

Adaptive Threshold Algorithm

1. **Boot Sampling (5s)**

   Collect 100 distance readings on startup to characterize the local environment.

2. **Dynamic Threshold Calc**

   WARNING threshold = baseline × 80%
   ALERT threshold = baseline × 50%

3. **Continuous Monitoring**

   Sample every 50ms. Each reading compared against computed thresholds.

4. **State Machine Dispatch**

   Zone transitions trigger ESP-NOW packet to Receiver node.

< 80ms

End-to-End Latency
Sensor → ESP-NOW → Actuator

50ms

Sampling Interval
HC-SR04 polling rate

80% / 50%

Adaptive Thresholds
Warning / Alert (× baseline)

5s

Boot Calibration
Environment characterization

Battery & Power Design

🔌 SENDER Device

USB-C Continuous Power

- Wall adapter or power bank via USB-C
- Ensures 24/7 uninterrupted 50ms sampling
- No battery anxiety — always-on monitoring
- Prioritizes sensor reliability over portability

🔋 RECEIVER Device

3.7V 820mAh LiPo Battery

- On-board XIAO charging IC (USB-C recharge)
- Portable deployment — place anywhere
- Low-voltage torque compensation in firmware
- ESP-NOW minimizes radio wake cycles

Battery Life Estimate

🔌 SENDER Device

USB-C Continuous Power

- 24/7

🔋 RECEIVER Device

3.7V 820mAh LiPo Battery

- 6-12 hours continuous use

Budget Summary

Component	Description	Unit Price	Qty	Total
XIAO ESP32-C3	Main MCU (×2)	$5.90	2	$11.80
HC-SR04	Ultrasonic distance sensor	$1.50	1	$1.50
X27 Stepper Motor	Mechanical actuator	$6.00	1	$6.00
Passive Buzzer	2.7kHz acoustic alert	$0.50	1	$0.50
5mm LED	Visual indicator	$0.20	2	$0.40
LiPo Battery 820mAh	Receiver node power	$8.00	1	$8.00
PCB Fabrication	Custom 2-layer PCB	$10.00	1	$10.00
Push Button Switch	Passive components	$5.00	1	$5.00
TOTAL				**$43.20**

Future Work

01 Computer Vision Integration

Replace ultrasonic distance alone with a camera + lightweight ML model (MobileNet) to distinguish cats from humans or other objects, reducing false positives.

02 Multi-Node Mesh Network

Extend ESP-NOW to support 5+ sensor nodes in a mesh topology, covering entire rooms or multi-floor apartments with a single coordinator.

03 Mobile App Dashboard

Build a companion iOS/Android app for real-time event logs, zone configuration, and battery status monitoring via BLE.

04 Deep Sleep Power Optimization

Implement wake-on-trigger deep sleep for Receiver node to extend battery life from days to weeks on a single charge.

05 Learned Behavior Profiling

Train a time-series model on intrusion patterns to predict and preemptively alert before threshold is crossed.

06 Industrial Packaging

Redesign into a consumer-grade enclosure with magnetic mounting and IP44 splash resistance for kitchen/outdoor use.